

These notes follow more or less directly from the Newey and McFadden (1994) handbook chapter, but I am writing out notes to verify I have all the formula exactly correct (mostly N s). I'm trying to write out all the steps to end up with the IF given in (6.6)

The first-stage QLD estimator can be written as a set of moments:

$$\mathbb{E} \left[\frac{D_{i\infty}}{\mathbb{P}[D_{i\infty} = 1]} \mathbf{H}(\boldsymbol{\theta}) \mathbf{y}_i \otimes \mathbf{w}_i \right] = \mathbf{0}, \quad (1)$$

where $\boldsymbol{\theta}$ is a vector of dimension $(T - p) * p$. The “quasi-long differencing” matrix $\mathbf{H}(\boldsymbol{\theta})$ takes the form:

$$\mathbf{H}(\boldsymbol{\theta}) = (\mathbf{I}_{T-p}, \boldsymbol{\Theta}). \quad (2)$$

For computation efficiency, it is helpful to expand $\mathbf{H}(\boldsymbol{\theta}) \mathbf{y}_i$ to

$$\mathbf{H}(\boldsymbol{\theta}) \mathbf{y}_i = y_{i,1:(T-p)} + \boldsymbol{\Theta} y_{i,(T-p+1):T}.$$

For each instrument $w_{i,l}$, there are $(T - p)$ moments implied by (1).

We estimate the following GMM problem using a two-stage estimation procedure:

$$\begin{aligned} Q_{1,N} &= \bar{\mathbf{m}}(\boldsymbol{\theta})' \tilde{\Omega}^{-1} \bar{\mathbf{m}}(\boldsymbol{\theta}) \\ &= \left(\frac{1}{N} \sum_i \mathbf{m}_i(\boldsymbol{\theta}) \right)' \left(\frac{1}{N} \sum_i \mathbf{m}_i(\tilde{\boldsymbol{\theta}}) \mathbf{m}_i(\tilde{\boldsymbol{\theta}})' \right)^{-1} \left(\frac{1}{N} \sum_i \mathbf{m}_i(\boldsymbol{\theta}) \right) \end{aligned}$$

We estimate the weights matrix, $\tilde{\Omega}$, using $\tilde{\boldsymbol{\theta}}$ from a first stage estimate where the identity matrix is used as the weighting matrix.

This produces the following first-order conditions

$$\left(\frac{1}{N} \sum_i \frac{\partial}{\partial \boldsymbol{\theta}} \mathbf{m}_i(\hat{\boldsymbol{\theta}}) \right)' \tilde{\Omega}^{-1} \left(\frac{1}{N} \sum_i \mathbf{m}_i(\hat{\boldsymbol{\theta}}) \right) \equiv \bar{\mathbf{M}}(\hat{\boldsymbol{\theta}})' \tilde{\Omega}^{-1} \bar{\mathbf{m}}(\hat{\boldsymbol{\theta}}) = \mathbf{0}$$

Expanding this at $\boldsymbol{\theta}_0$ gives (asymptotically)

$$\mathbf{0} = \bar{\mathbf{M}}(\hat{\boldsymbol{\theta}})' \tilde{\Omega}^{-1} \bar{\mathbf{m}}(\hat{\boldsymbol{\theta}}) = \bar{\mathbf{M}}(\hat{\boldsymbol{\theta}})' \tilde{\Omega}^{-1} \bar{\mathbf{m}}(\boldsymbol{\theta}_0) + \bar{\mathbf{M}}(\hat{\boldsymbol{\theta}})' \tilde{\Omega}^{-1} \bar{\mathbf{M}}(\bar{\boldsymbol{\theta}}) (\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0)$$

Solving and multiplying by $\sqrt{N}$ yields

$$\begin{aligned}\sqrt{N}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) &= - \left(\bar{M}(\hat{\boldsymbol{\theta}})' \tilde{\Omega}^{-1} \bar{M}(\bar{\boldsymbol{\theta}}) \right)^{-1} \bar{M}(\hat{\boldsymbol{\theta}})' \tilde{\Omega}^{-1} \sqrt{N} \bar{m}(\boldsymbol{\theta}_0) \\ &= - \left[\left(\frac{1}{N} \sum_i \frac{\partial}{\partial \boldsymbol{\theta}} m_i(\hat{\boldsymbol{\theta}}) \right)' \left(\frac{1}{N} \sum_i m_i(\tilde{\boldsymbol{\theta}}) m_i(\tilde{\boldsymbol{\theta}})' \right) \left(\frac{1}{N} \sum_i \frac{\partial}{\partial \boldsymbol{\theta}} m_i(\bar{\boldsymbol{\theta}}) \right) \right]^{-1} \times \\ &\quad \left(\frac{1}{N} \sum_i \frac{\partial}{\partial \boldsymbol{\theta}} m_i(\hat{\boldsymbol{\theta}}) \right)' \left(\frac{1}{N} \sum_i m_i(\tilde{\boldsymbol{\theta}}) m_i(\tilde{\boldsymbol{\theta}})' \right) \left(\sqrt{N} \frac{1}{N} \sum_i m_i(\boldsymbol{\theta}_0) \right)\end{aligned}$$

The previous display gives the influence function for the first-step quasi-long differencing estimator. The first five terms converge (under regularity conditions) to non-random matrices. The last term will, by some CLT, converge in distribution to a normally distributed term.

Then, our treatment effect estimator can be written as a method of moments estimator given by

$$g(z_i, \boldsymbol{\tau}, \hat{\boldsymbol{\theta}}) \equiv g_i(\boldsymbol{\tau}, \hat{\boldsymbol{\theta}}),$$

where $\hat{\boldsymbol{\theta}}$ is the first-step estimate. The method of moments estimator solves the sample average of g_i :

$$\frac{1}{N} \sum_i g_i(\hat{\boldsymbol{\tau}}, \hat{\boldsymbol{\theta}}) = \mathbf{0}$$

Expanding this at the true value $\boldsymbol{\tau}_0$ yields

$$\mathbf{0} = \frac{1}{N} \sum_i g_i(\hat{\boldsymbol{\tau}}, \hat{\boldsymbol{\theta}}) = \frac{1}{N} \sum_i g_i(\boldsymbol{\tau}_0, \hat{\boldsymbol{\theta}}) + \left[\frac{1}{N} \sum_i \frac{\partial}{\partial \boldsymbol{\tau}} g_i(\bar{\boldsymbol{\tau}}, \hat{\boldsymbol{\theta}}) \right] (\hat{\boldsymbol{\tau}} - \boldsymbol{\tau}_0)$$

We can solve and multiply by $\sqrt{N}$ to yield

$$\sqrt{N}(\hat{\boldsymbol{\tau}} - \boldsymbol{\tau}_0) = - \left[\frac{1}{N} \sum_i \frac{\partial}{\partial \boldsymbol{\tau}} g_i(\bar{\boldsymbol{\tau}}, \hat{\boldsymbol{\theta}}) \right]^{-1} \frac{1}{\sqrt{N}} \sum_i g_i(\boldsymbol{\tau}_0, \hat{\boldsymbol{\theta}})$$

Expanding $\frac{1}{\sqrt{N}} \sum_i g_i(\tau_0, \hat{\theta})$ at θ_0 yields

$$\begin{aligned} \frac{1}{\sqrt{N}} \sum_i g_i(\tau_0, \hat{\theta}) &= \frac{1}{\sqrt{N}} \sum_i g_i(\tau_0, \theta_0) + \left[\frac{1}{N} \sum_i \frac{\partial}{\partial \tau} g_i(\tau_0, \bar{\theta}) \right] \sqrt{N}(\hat{\theta} - \theta_0) \\ &= \frac{1}{\sqrt{N}} \sum_i g_i(\tau_0, \theta_0) + \\ &\quad \left[\frac{1}{N} \sum_i \frac{\partial}{\partial \theta} g_i(\tau_0, \bar{\theta}) \right] \left(\bar{M}(\hat{\theta})' \tilde{\Omega}^{-1} \bar{M}(\bar{\theta}) \right)^{-1} \bar{M}(\hat{\theta})' \tilde{\Omega}^{-1} \sqrt{N} \bar{m}(\theta_0) \end{aligned}$$

Combining the previous two displays yield the influence function for our treatment effect estimator:

$$\begin{aligned} \sqrt{N}(\hat{\tau} - \tau_0) &= \left[\frac{1}{N} \sum_i \frac{\partial}{\partial \tau} g_i(\bar{\tau}, \hat{\theta}) \right]^{-1} \frac{1}{\sqrt{N}} \sum_i g_i(\tau_0, \theta_0) + \\ &\quad \left[\frac{1}{N} \sum_i \frac{\partial}{\partial \tau} g_i(\bar{\tau}, \hat{\theta}) \right]^{-1} \left[\frac{1}{N} \sum_i \frac{\partial}{\partial \theta} g_i(\tau_0, \bar{\theta}) \right] \left(\bar{M}(\hat{\theta})' \tilde{\Omega}^{-1} \bar{M}(\bar{\theta}) \right)^{-1} \times \\ &\quad \bar{M}(\hat{\theta})' \tilde{\Omega}^{-1} \left(\frac{1}{\sqrt{N}} \sum_i m_i(\theta_0) \right). \end{aligned}$$

Here, the first term is the standard OLS score and the second term adjusts for the fact that θ is estimated.